

Lab 12.1: Incident Response Procedures, Forensics, and Forensics Analysis

Lab 12.1: Part 1 – Launching an Attack

1. Question 3

```
C:\>nmap 203.0.113.100

Starting Nmap 5.51 ( http://nmap.org ) at 2025-11-02 17:51 Eastern Standard Time
Nmap scan report for campus.edu (203.0.113.100)
Host is up (0.00s latency).
Not shown: 989 filtered ports
PORT      STATE SERVICE
21/tcp    open  ftp
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
110/tcp   open  pop3
443/tcp   open  https
1099/tcp  open  rmiregistry
3306/tcp  open  mysql
3389/tcp  open  ms-term-serv
5432/tcp  open  postgresql
8180/tcp  open  sampleflag:999818

Nmap done: 1 IP address (1 host up) scanned in 12.66 seconds
C:\>_
```

2. Question 5

Port	Protocol	State	Service	Version
21	tcp	open	ftp	Microsoft ftpd
23	tcp	open	telnet	
25	tcp	open	smtp	hMailServer smtpd
80	tcp	open	http	Apache httpd 2.2.14 ((Win32) DAV/2 mod_ssl/2.2.14
110	tcp	open	pop3	hMailServer pop3d
443	tcp	open	http	Apache httpd 2.2.14 ((Win32) DAV/2 mod_ssl/2.2.14
1099	tcp	open	rmi	Java RMI Registry
3306	tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
3389	tcp	open	microsoft-rdp	Microsoft Terminal Service
5432	tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
8180	tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1

3. Question 13

Result Testing Message				
Target	Service	User	Password	N
203.0.113.100	ftp	administrator	P@ssw0rd	

4. Question 15

```
C:\>nmap 203.0.113.100 -p 3389

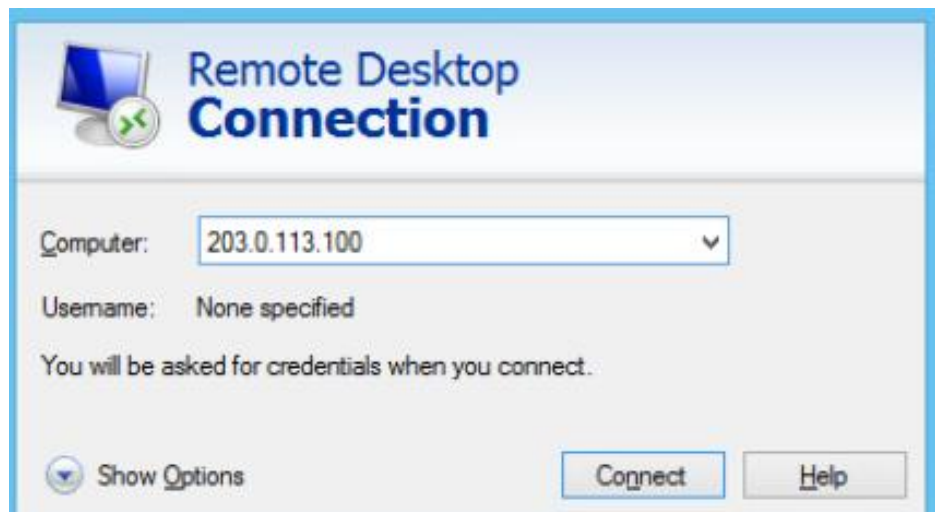
Starting Nmap 5.51 ( http://nmap.org ) at 2025-11-02 18:26 Eastern Standard Time
Nmap scan report for campus.edu (203.0.113.100)
Host is up (0.00s latency).
PORT      STATE SERVICE
3389/tcp  open  ms-term-serv

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
C:\>
```

5. Question 17

**Lab
12.1:
Part 2**

—



Collecting Incident Response Data

1. Question 8

```
c E:\>PATH=E:\
io E:\>
```

2. Question 14

```
c E:\>PATH=E:\
io E:\>
```

3. Question 16

```
E:\>type IR.txt
incident Responder student
Sun 11/02/2025

Windows IP Configuration

Host Name . . . . . : SERVER
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection 3:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 MT Network Conne
Physical Address. . . . . : 00-50-56-02-47-C0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8c66:b98f:ed29:c58a%13(Prefe
IPv4 Address. . . . . : 192.168.1.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.254
DNS Servers . . . . . : 192.168.1.10
```

4. Question 18

```
E:\>netstat -ano >> IR.txt

E:\>type IR.txt | find "3389"
TCP        0.0.0.0:3389          0.0.0.0:0             LISTENING
TCP        [::]:3389             [::]:0                 LISTENING
```

5. Question 20

```
E:\>type IR.txt | find "svchost"
svchost.exe      856 Services          0
svchost.exe      924 Services          0
svchost.exe     1004 Services          0
svchost.exe     1060 Services          0
svchost.exe     1076 Services          0
svchost.exe     1148 Services          0
svchost.exe     1200 Services          0
svchost.exe     1252 Services          0
svchost.exe     1388 Services          0
svchost.exe     1240 Services          0
svchost.exe     1020 Services          0
svchost.exe     2240 Services          0
svchost.exe     4088 Services          0
```

6. Question 22

```
E:\>type IR.txt | find "administrator"
11/2/2025 6:29:04 PM      CAMPUS\administrator
```

7. Question 24

```
E:\>type IR.txt | find "Guest"
Guest                  IUSR_WINFILE          krbtgt
```

8. Question 27

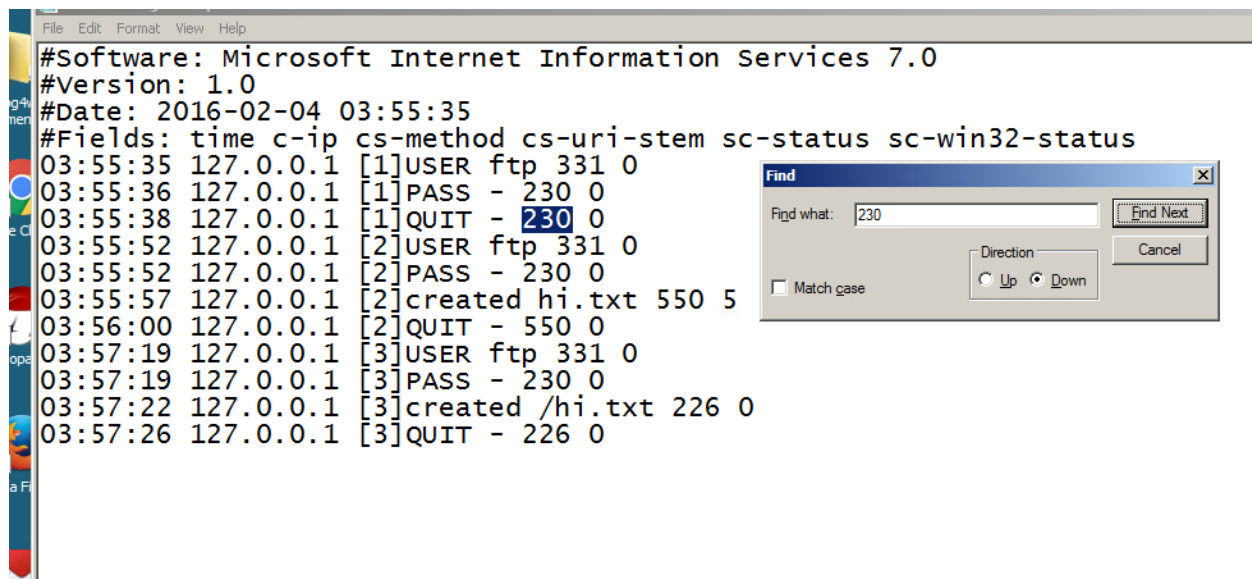
```

E:\>type IR.txt | find "Guest"
Guest                                IUSR_WINFILE                        krbtgt

```

Lab 12.1: Part 3 – Viewing Logs

1. Question 9



Lab 12.1: Part 4 – Lessons Learned

A primary lesson learned is the severe lack of security hardening and vulnerability management. The initial network scan revealed a highly exposed host with numerous open and potentially outdated services, including FTP (21/tcp), Telnet (23/tcp), HTTP (80/tcp), and Remote Desktop Services (3389/tcp). The presence of Telnet is particularly concerning as it transmits credentials in clear text, and the sheer number of exposed ports significantly widens the attack surface of the system, which was identified as a server